SR UNIVERSITY

School of Computer Science and Artificial Intelligence

Lab Assignment # 14

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Task Description 1 – (Landing Page Design)

- **Create a simple landing page for a product or service. Include a header, hero section, and footer.**

Requirements:

- **Use HTML and CSS to structure the page.**
- **Use AI to suggest layout alignment and typography.**
- **Ensure the page is responsive across devices.**

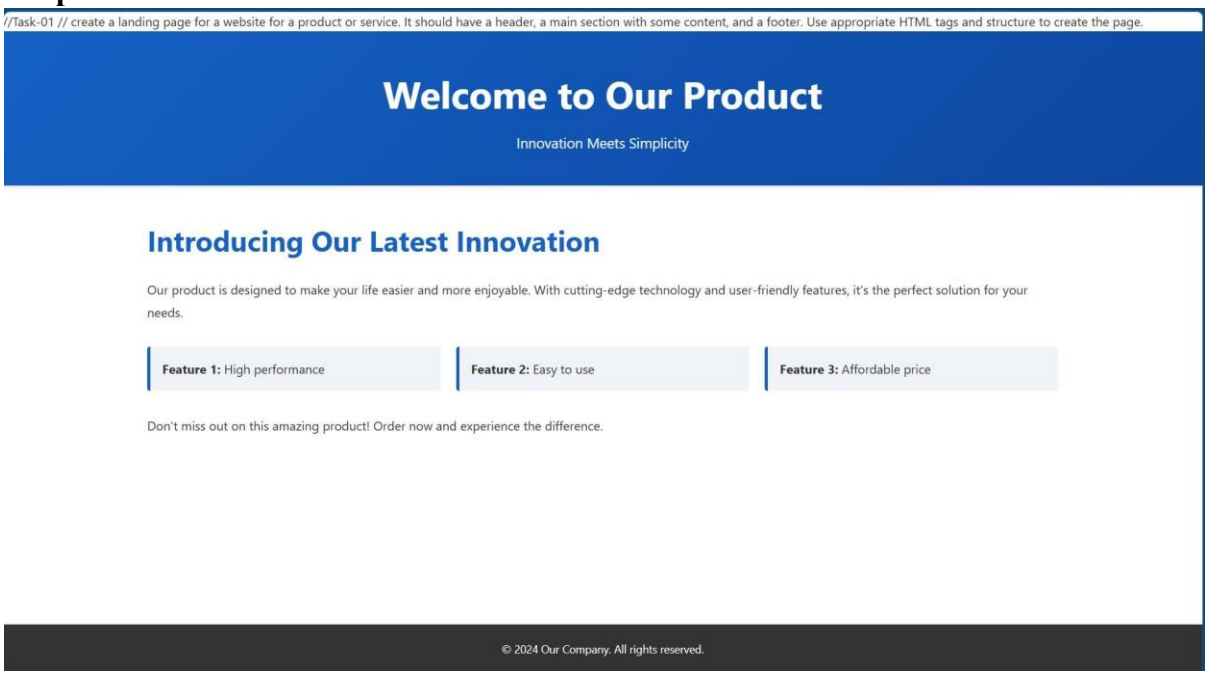
Prompt:create a landing page for a website for a product or service. It should have a header, a main section with some content, and a footer. Use appropriate HTML tags and structure to create the page. **Code:-**

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Product Landing Page</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            line-height: 1.6;
            color: #333;
        }
        header {
            background: linear-gradient(135deg, #1561c4 0%, #0d47a1 100%);
            color: white;
            padding: 40px 20px;
            text-align: center;
            box-shadow: 0 2px 5px rgba(0,0,0,0.1);
        }
        header h1 {
            font-size: clamp(24px, 5vw, 48px);
            margin-bottom: 10px;
        }
        header p {
            font-size: clamp(14px, 2vw, 18px);
            opacity: 0.9;
        }
        main {
            padding: 40px 20px;
            max-width: 1200px;
            margin: 0 auto;
            padding-bottom: 120px;
        }
        main h2 {
            font-size: clamp(20px, 4vw, 36px);
            margin-bottom: 20px;
            color: #1561c4;
```

```css
}
main p {
    font-size: clamp(14px, 1.5vw, 16px);
    margin-bottom: 20px;
    line-height: 1.8;
}
ul {
    list-style: none;
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
    gap: 20px;
    margin: 30px 0;
}
li {
    background: #f0f4f8;
    padding: 15px;
    border-left: 4px solid #1561c4;
    border-radius: 4px;
}
footer {
    background-color: #333;
    color: white;
    text-align: center;
    padding: 20px;
    position: fixed;
    width: 100%;
    bottom: 0;
    font-size: clamp(12px, 1vw, 14px);
}
@media (max-width: 768px) {
    main {
        padding: 30px 15px;
        padding-bottom: 100px;
    }
    ul {
        grid-template-columns: 1fr;
    }
}
```

```html
    </style>
</head>
<body>
    <header>
        <h1>Welcome to Our Product</h1>
        <p>Innovation Meets Simplicity</p>
    </header>
    <main>
        <h2>Introducing Our Latest Innovation</h2>
        <p>Our product is designed to make your life easier and more enjoyable. With cutting-edge technology and user-friendly features,
        <ul>
            <li><strong>Feature 1:</strong> High performance</li>
            <li><strong>Feature 2:</strong> Easy to use</li>
            <li><strong>Feature 3:</strong> Affordable price</li>
        </ul>
        <p>Don't miss out on this amazing product! Order now and experience the difference.</p>
    </main>
    <footer>
        &copy; 2024 Our Company. All rights reserved.
    </footer>
</body>
</html>
```

Output:-



Justification:-
Showcases a product section with clear headings and bullet points to highlight key features and benefits for users. Focuses on readable layout and persuasive copy to improve user engagement and conversion.

Task Description 2 – (Theme Toggle Feature)
• Provide a button to switch between light and dark themes..
Requirements:
• Use JavaScript to toggle CSS classes.
• Use AI assistance to manage color contrast and accessibility.
• Ensure theme preference applies instantly on click.

Prompt:create a button that can switch light and dark themes using javascript to toggle css classes and to manage color contrast and accessibility to ensure theme perfomance instanly on clicking.

Code:-

```html
<html>
    <head>
        <style>
            body.light-theme {
            }
            body.dark-theme {
                background-color: #222222;
                color: #ffffff;
            }
            .theme-toggle-btn {
                margin: 20px;
                padding: 10px 20px;
                font-size: 1.2em;
                cursor: pointer;
                border: none;
                border-radius: 5px;
                background: #e0e0e0;
                color: #222;
                transition: background 0.2s, color 0.2s;
            }
            body.dark-theme .theme-toggle-btn {
                background: #444;
                color: #fff;
            }
        </style>
    </head>
    <body class="light-theme">
        <button id="themeToggle" class="theme-toggle-btn" aria-pressed="false" aria-label="Switch to dark theme">
            Switch to Dark Theme
        </button>
        <script>
            const btn = document.getElementById('themeToggle');
            const body = document.body;
            function updateAccessibility(isDark) {
                btn.setAttribute('aria-pressed', isDark);
                btn.setAttribute('aria-label', isDark ? 'Switch to light theme' : 'Switch to dark theme');
                btn.textContent = isDark ? 'Switch to Light Theme' : 'Switch to Dark Theme';
            }
            btn.addEventListener('click', () => {
                const isDark = body.classList.toggle('dark-theme');
                body.classList.toggle('light-theme', !isDark);
                updateAccessibility(isDark);
            });
            // Set correct theme on load if preferring dark mode
            const prefersDark = window.matchMedia('(prefers-color-scheme: dark)').matches;
            if (prefersDark) {
                body.classList.remove('light-theme');
                body.classList.add('dark-theme');
                updateAccessibility(true);
            }
        </script>
    </body>
</html>
```

Output:-

In white Theme

In Black Theme

Justification:-
Implements a theme toggle button that instantly switches between light and dark modes using class-based styling. Improves accessibility and user comfort by adjusting colors dynamically without reloading the page.

Task Description 3 – (User Feedback Form)
• Create a feedback form with fields: Username, Rating, Comments.
Requirements:
• Use JavaScript to validate required fields.
• Display success or error messages dynamically.
• Use AI to improve form usability and layout.

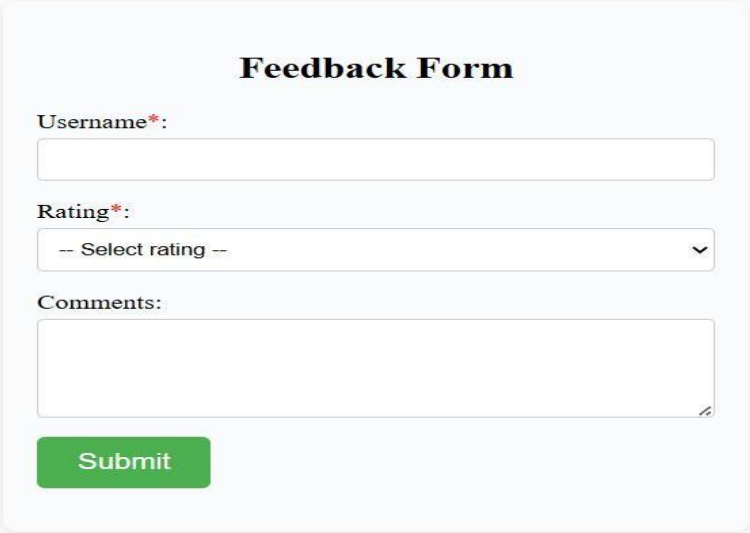
Prompt:create a feedback form with info of username, rating, comments using javascript to validate required field and usabilty and layout.

Code:-

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Feedback Form</title>
</head>
<body>
  <div id="formContainer"></div>
  <script>
    // Create the form HTML structure dynamically
    const formContainer = document.getElementById('formContainer');
    formContainer.innerHTML = `
      <form id="feedbackForm" style="max-width:400px;margin:30px auto;padding:20px;border-radius:8px;box-shadow:0 2px 8px #ddd;background:#fafbfc;">
        <h2 style="text-align:center;">Feedback Form</h2>
        <div style="margin-bottom:15px;">
          <label for="username" style="display:block;margin-bottom:4px;">Username<span style="color:red;">*</span>:</label>
          <input type="text" id="username" name="username" required style="width:100%;padding:8px;border-radius:4px;border:1px solid #ccc;">
          <div id="usernameError" style="color:red;font-size:0.95em;display:none;margin-top:2px;"></div>
        </div>
        <div style="margin-bottom:15px;">
          <label for="rating" style="display:block;margin-bottom:4px;">Rating<span style="color:red;">*</span>:</label>
          <select id="rating" name="rating" required style="width:100%;padding:8px;border-radius:4px;border:1px solid #ccc;">
            <option value="">-- Select rating --</option>
            <option value="5">5 - Excellent</option>
            <option value="4">4 - Good</option>
            <option value="3">3 - Average</option>
            <option value="2">2 - Poor</option>
            <option value="1">1 - Terrible</option>
          </select>
          <div id="ratingError" style="color:red;font-size:0.95em;display:none;margin-top:2px;"></div>
        </div>
        <div style="margin-bottom:15px;">
          <label for="comments" style="display:block;margin-bottom:4px;">Comments:</label>
          <textarea id="comments" name="comments" rows="4" style="width:100%;padding:8px;border-radius:4px;border:1px solid #ccc;"></textarea>
        </div>
        <button type="submit" style="padding:10px 24px;border:none;border-radius:5px;background:#4CAF50;color:#fff;font-size:1.1em;cursor:pointer;">Submit</button>
        <div id="formFeedback" style="margin-top:15px;font-weight:bold;"></div>
      </form>
    `;
    // Form validation and usability
```

```
    const feedbackForm = document.getElementById('feedbackForm');
    const usernameInput = document.getElementById('username');
    const ratingInput = document.getElementById('rating');
    const commentsInput = document.getElementById('comments');
    const usernameError = document.getElementById('usernameError');
    const ratingError = document.getElementById('ratingError');
    const formFeedback = document.getElementById('formFeedback');
    function validateUsername() {
      if (usernameInput.value.trim() === "") {
        usernameError.textContent = "Username is required.";
        usernameError.style.display = "block";
        return false;
      } else {
        usernameError.textContent = "";
        usernameError.style.display = "none";
        return true;
      }
    }
    function validateRating() {
      if (!ratingInput.value) {
        ratingError.textContent = "Please select a rating.";
        ratingError.style.display = "block";
        return false;
      } else {
        ratingError.textContent = "";
        ratingError.style.display = "none";
        return true;
      }
    }
    // Accessibility: show errors live
    usernameInput.addEventListener('input', validateUsername);
    ratingInput.addEventListener('change', validateRating);
    feedbackForm.addEventListener('submit', function (e) {
      e.preventDefault();
      const isUsernameValid = validateUsername();
      const isRatingValid = validateRating();
      if (isUsernameValid && isRatingValid) {
        formFeedback.style.color = "green";
        formFeedback.textContent = "Thank you for your feedback!";
        feedbackForm.reset();
        usernameError.style.display = "none";
        ratingError.style.display = "none";
        setTimeout(() => {formFeedback.textContent = "";}, 3000);
      } else {
        formFeedback.style.color = "red";
        formFeedback.textContent = "Please correct the errors above and submit again.";
      }
    });
  </script>
</body>
</html>
```

Output:-

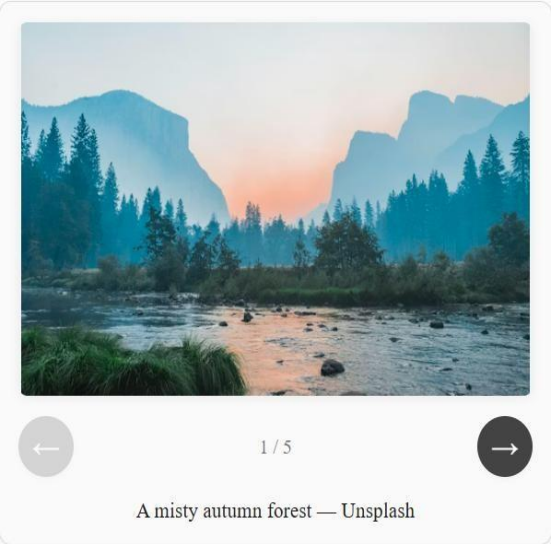


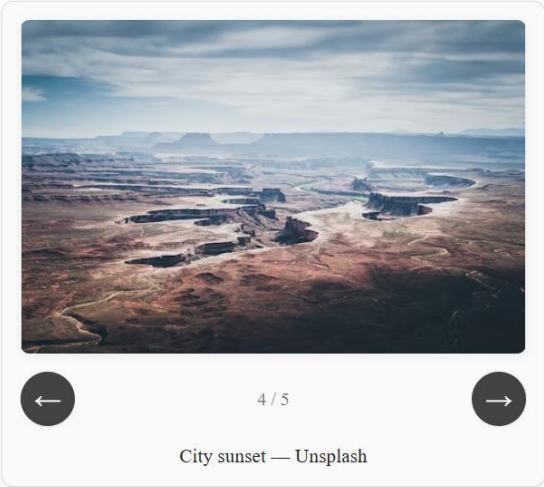
Justifications:-
Builds a feedback form capturing username, rating, and comments with JavaScript validation for required fields. Enhances usability by preventing incomplete submissions and guiding users to correct input errors.

Task Description - 4 (Image Gallery with JavaScript)
• Task: Display a small image gallery on a web page.
Requirements:
• Use JavaScript to dynamically load images.
• Add next/previous navigation buttons.
• Use AI to optimize layout responsiveness.

Prompt:create a small image gallery on a webpage using html,js to dynamically load images and add navigations on next/previous buttons and optimize layout responsiveness, HTML structure for the image gallery

Output:-





Justification:-

Creates a responsive image gallery that dynamically loads images and supports next/previous navigation buttons. Provides a clean, interactive browsing experience optimized for different screen sizes

Code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Responsive JavaScript Image Gallery</title>
  <style>
    :root {
      --gallery-max-width: 480px;
      --gallery-margin: 0 auto;
      --gallery-bg: #fafafa;
      --gallery-border: 1px solid #ddd;
      --gallery-radius: 10px;
      --btn-bg: #333;
      --btn-color: #fff;
      --btn-radius: 50%;
      --btn-size: 48px;
    }

    .gallery-container {
      max-width: var(--gallery-max-width);
      margin: var(--gallery-margin);
      background: var(--gallery-bg);
      border: var(--gallery-border);
      border-radius: var(--gallery-radius);
      padding: 16px;
      box-sizing: border-box;
      display: flex;
      flex-direction: column;
      align-items: center;
      box-shadow: 0 2px 10px 0 rgba(0,0,0,0.08);
    }
    .gallery-image-wrapper {
      position: relative;
      width: 100%;
      padding-top: 66%; /* Optimized for 3:2 ratio fo
      background: #f1f2f4;
      border-radius: 8px;
      overflow: hidden;
      margin-bottom: 16px;
      transition: box-shadow .18s;
      box-shadow: 0 2px 8px 0 rgba(0,0,0,.06);
      display: flex;
      align-items: center;
      justify-content: center;
    }
    .gallery-image-wrapper img {
      margin: 0 auto;
      position: absolute;
      left: 0; top: 0; width: 100%; height: 100%;
      object-fit: contain;
      transition: opacity .16s;
      user-select: none;
      pointer-events: none;
    }
```

```css
.gallery-buttons {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 8px;
}
.gallery-btn {
  background: var(--btn-bg);
  color: var(--btn-color);
  border: none;
  border-radius: var(--btn-radius);
  width: var(--btn-size);
  height: var(--btn-size);
  font-size: 1.5em;
  cursor: pointer;
  transition: background .18s, filter .18s;
  display: flex;
  align-items: center;
  justify-content: center;
  outline: none;
  opacity: 0.92;
  box-shadow: 0 2px 4px #acacac22;
  z-index: 2;
}
.gallery-btn:disabled {
  background: #bbb;
  cursor: not-allowed;
  opacity: 0.6;
}

.gallery-caption {
  font-size: 1.07em;
  text-align: center;
  margin-top: 0.5em;
  color: #222;
}
@media (max-width: 600px) {
  .gallery-container {
    padding: 8px;
    max-width: 98vw;
  }
  .gallery-image-wrapper {
    padding-top: 60%;
    border-radius: 6px;
  }
  .gallery-btn {
    width: 40px;
    height: 40px;
    font-size: 1.1em;
  }
  .gallery-caption {
    font-size: 0.97em;
  }
}
@media (max-width: 400px) {
  .gallery-btn {
    width: 36px;
    height: 36px;
    font-size: 0.95em;
  }
  .gallery-image-wrapper {
    padding-top: 56%;
  }
```

-

```html
</style>
</head>
<body>
<div class="gallery-container">
  <div class="gallery-image-wrapper" id="imgWrapper">
    <!-- The image will be inserted here by JS -->
  </div>
  <div class="gallery-buttons">
    <button class="gallery-btn" id="prevBtn" aria-label="Previous image">&larr;</button>
    <span id="galleryCounter" style="min-width: 50px;text-align: center;font-size: 1em; color: #777"></span>
    <button class="gallery-btn" id="nextBtn" aria-label="Next image">&rarr;</button>
  </div>
  <div class="gallery-caption" id="imgCaption"></div>
</div>
<script>
  // AI optimized: Use small, modern placeholder pics & alt text for a11y
  const images = [
    {
      src: "https://images.unsplash.com/photo-1506744038136-46273834b3fb?auto=format&fit=crop&w=600&q=80",
      alt: "A forest in autumn morning mist",
      caption: "A misty autumn forest — Unsplash"
    },
    {
      src: "https://images.unsplash.com/photo-1519125323398-675f0ddb6308?auto=format&fit=crop&w=600&q=80",
      alt: "Snowy mountain peak with clouds",
      caption: "Snowy mountain — Unsplash"
    },
    {
      src: "https://images.unsplash.com/photo-1465101046530-73398c7f28ca?auto=format&fit=crop&w=600&q=80",
      alt: "Winding road in green landscape",
      caption: "Green landscape road — Unsplash"
    },
    {
      src: "https://images.unsplash.com/photo-1519985176271-adb1088fa94c?auto=format&fit=crop&w=600&q=80",
      alt: "City skyline at sunset",
      caption: "City sunset — Unsplash"
    },
    {
      src: "https://images.unsplash.com/photo-1502082553048-f009c37129b9?auto=format&fit=crop&w=600&q=80",
      alt: "Closeup of sunflower",
      caption: "Sunflower closeup — Unsplash"
    }
  ];

  let currentIdx = 0;

  const imgWrapper = document.getElementById('imgWrapper');
  const prevBtn = document.getElementById('prevBtn');
  const nextBtn = document.getElementById('nextBtn');
  const imgCaption = document.getElementById('imgCaption');
  const galleryCounter = document.getElementById('galleryCounter');
```

```cpp
function updateGallery(idx) {
  // Remove previous img
  imgWrapper.innerHTML = "";
  // Responsive: use srcset for layout and bandwidth optimization
  const img = document.createElement('img');
  img.src = images[idx].src;
  img.alt = images[idx].alt;
  img.decoding = "async";
  img.loading = "lazy";
  img.srcset = images[idx].src.replace("w=600", "w=400") + " 400w, " +
               images[idx].src + " 600w";
  img.sizes = "(max-width: 480px) 98vw, 480px";
  imgWrapper.appendChild(img);

  imgCaption.textContent = images[idx].caption;
  galleryCounter.textContent = (idx+1) + " / " + images.length;

  prevBtn.disabled = idx === 0;
  nextBtn.disabled = idx === images.length - 1;
}
// Handle navigation
prevBtn.addEventListener('click', () => {
  if (currentIdx > 0) {
    currentIdx--;
    updateGallery(currentIdx);
  }
});
nextBtn.addEventListener('click', () => {
  if (currentIdx < images.length - 1) {
    currentIdx++;
    updateGallery(currentIdx);
  }
});

// Keyboard accessibility
document.addEventListener('keydown', e => {
  if (e.key === "ArrowLeft" && currentIdx > 0) {
    currentIdx--;
    updateGallery(currentIdx);
  }
  if (e.key === "ArrowRight" && currentIdx < images.length - 1) {
    currentIdx++;
    updateGallery(currentIdx);
  }
});

// Initial load
updateGallery(currentIdx);
</script>
</body>
</html>
```

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Task Description - 5: (Notification Banner) • Display a notification banner at the top of the page.
Requirements:
• Use JavaScript to show and hide the banner.
• Style the banner using CSS.
• Use AI to ensure smooth transitions and clean UI behavior.

Prompt:create a webpage to dispaly the notification banner at th e top of the page using html,js to show the hidden the bannerand style the banner using css to ensure smooth transition and clean UI behaviour.

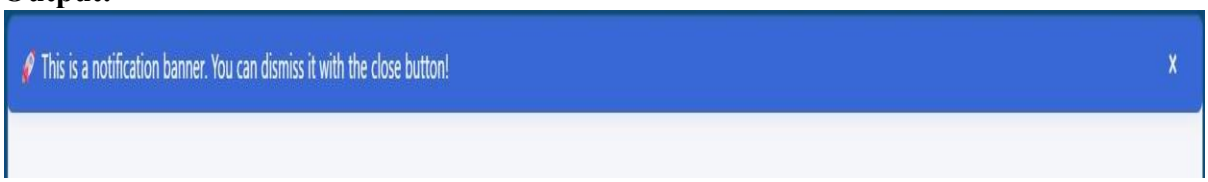
Code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Notification Banner Example</title>
  <style>
    :root {
      --banner-bg: #3867d6;
      --banner-color: #fff;
      --banner-height: 48px;
      --banner-font: 1.05em;
      --banner-radius: 0 0 7px 7px;
      --banner-shadow: 0 2px 12px rgba(44,62,80,0.08);
      --banner-transition: transform 0.35s cubic-bezier(.8,.22,.23,.89), opacity .25s;
      --close-btn-size: 28px;
      --close-btn-bg: transparent;
      --close-btn-hover: #fff5;
      --close-btn-color: #fff;
    }
    body {
      margin: 0;
      font-family: system-ui,sans-serif;
      background: #f5f6fa;
      min-height: 400px;
    }
    .notif-banner {
      box-sizing: border-box;
      width: 100%;
      background: var(--banner-bg);
      color: var(--banner-color);
      min-height: var(--banner-height);
      display: flex;
      align-items: center;
      justify-content: space-between;
      position: fixed;
      left: 0;
      top: 0;
      z-index: 1000;
      border-radius: var(--banner-radius);
      box-shadow: var(--banner-shadow);
      padding: 0 24px 0 16px;
      font-size: var(--banner-font);
      letter-spacing: 0.01em;
      transform: translateY(-110%);
      opacity: 0;
      pointer-events: none;
      transition: var(--banner-transition);
    }
    .notif-banner.show {
      transform: translateY(0);
      opacity: 1;
      pointer-events: auto;
    }
```

```html
    .notif-banner__text {
      flex: 1;
      padding: 6px 0;
      text-align: left;
    }
    .notif-banner__close {
      appearance: none;
      border: none;
      background: var(--close-btn-bg);
      color: var(--close-btn-color);
      border-radius: 50%;
      width: var(--close-btn-size);
      height: var(--close-btn-size);
      font-size: 1.22em;
      cursor: pointer;
      display: flex;
      align-items: center;
      justify-content: center;
      transition: background 0.18s;
      outline: none;
      margin-left: 14px;
    }
    .notif-banner__close:focus,
    .notif-banner__close:hover {
      background: var(--close-btn-hover);
    }

    /* Display 'Show' button only when bann
    #showBannerBtn {
      position: fixed;
      top: 14px;
      right: 16px;
      z-index: 900;
      padding: 7px 16px;
      background: #fff;
      border: 1px solid #3867d6;
      color: #3867d6;
      border-radius: 18px;
      font-size: 1em;
      font-weight: 500;
      cursor: pointer;
      box-shadow: 0 0 8px #3867d610;
      opacity: 0;
      pointer-events: none;
      transition: opacity 0.22s;
    }
    #showBannerBtn.active {
      opacity: 1;
      pointer-events: auto;
    }
    @media (max-width: 480px) {
      .notif-banner {
        font-size: .95em;
        padding-right: 8px;
      }
      #showBannerBtn {
        font-size: .97em;
        right: 8px;
      }
    }
  </style>
</head>
<body>
```

```html
<body>
  <div class="notif-banner" id="notifBanner" aria-live="polite" role="status">
    <span class="notif-banner__text">🚀 This is a notification banner. You can dismiss it with the close button!</span>
    <button class="notif-banner__close" id="closeBannerBtn" aria-label="Dismiss notification">&times;</button>
  </div>
  <button id="showBannerBtn">Show Notification</button>

  <script>
    // Elements
    const banner = document.getElementById('notifBanner');
    const closeBtn = document.getElementById('closeBannerBtn');
    const showBtn = document.getElementById('showBannerBtn');

    // Show the banner on page load (AI/UX: small delay to catch the eye)
    setTimeout(() => {
      banner.classList.add('show');
      showBtn.classList.remove('active');
    }, 250);

    // Hide the banner
    closeBtn.addEventListener('click', () => {
      banner.classList.remove('show');
      // Show the 'show' button for accessibility
      setTimeout(() => showBtn.classList.add('active'), 400);
    });

    // Show the banner again when user clicks 'Show Notification'
    showBtn.addEventListener('click', () => {
      banner.classList.add('show');
      showBtn.classList.remove('active');
    });

    // Optional: Hide banner on Escape key for a11y
    document.addEventListener('keydown', e => {
      if (e.key === "Escape" && banner.classList.contains('show')) {
        banner.classList.remove('show');
        setTimeout(() => showBtn.classList.add('active'), 400);
      }
    });
  </script>
</body>
</html>
```

Output:-



Justification:-

Displays a notification banner at the top of the page that can be shown or hidden using JavaScript. Uses smooth transitions and styled layout to deliver important messages without disrupting the overall UI.